PRACTICAL APPLICATION:

1.Alert classification system:

Alert classification system:

Detect alerts in wazuh , goto threat hunting → events

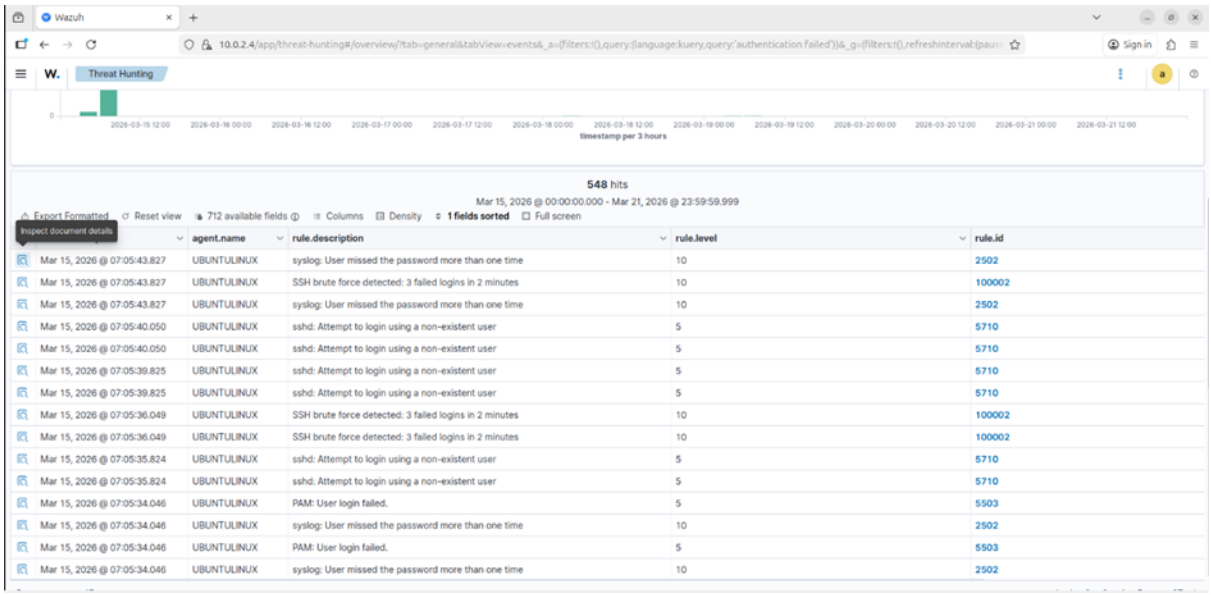

Figure 1: alerts in wazuh

Create a google sheet to map detected alerts to MITRE ATT&CK technique

Alert ID	Type	Priority	MITRE Tactic
5710	authentication failed	medium	T1110.001
2502	bruteforce	high	T1110
5503	password guessing	medium	T1110.001
60122	Account access removal	medium	T1531

Figure 2: Created a Google Sheets table to map alerts to MITRE ATT&CK Technique

Prioritize Alerts:
Prioritizing alerts using score using CVSS Score in Google Sheets

alert name	cvss	priority
ransomware detected	9.5	critical
SQL injection	9	critical
phishing email	7.5	high
malware detected	7.8	high
bruteforce login attack	6.5	medium
port scan	3.5	low
login from new location	3	low

Figure 3: prioritizing alerts using score using CVSS Score in Google Sheets

Dashboard Creation:
In wazuh creating dashboard to visualize the priorities

- Goto dashboard → create visualization → choose visualization type as piechart → choose index as wasuh.alerts-*
- Configure piechart
  Metric → aggregation : count

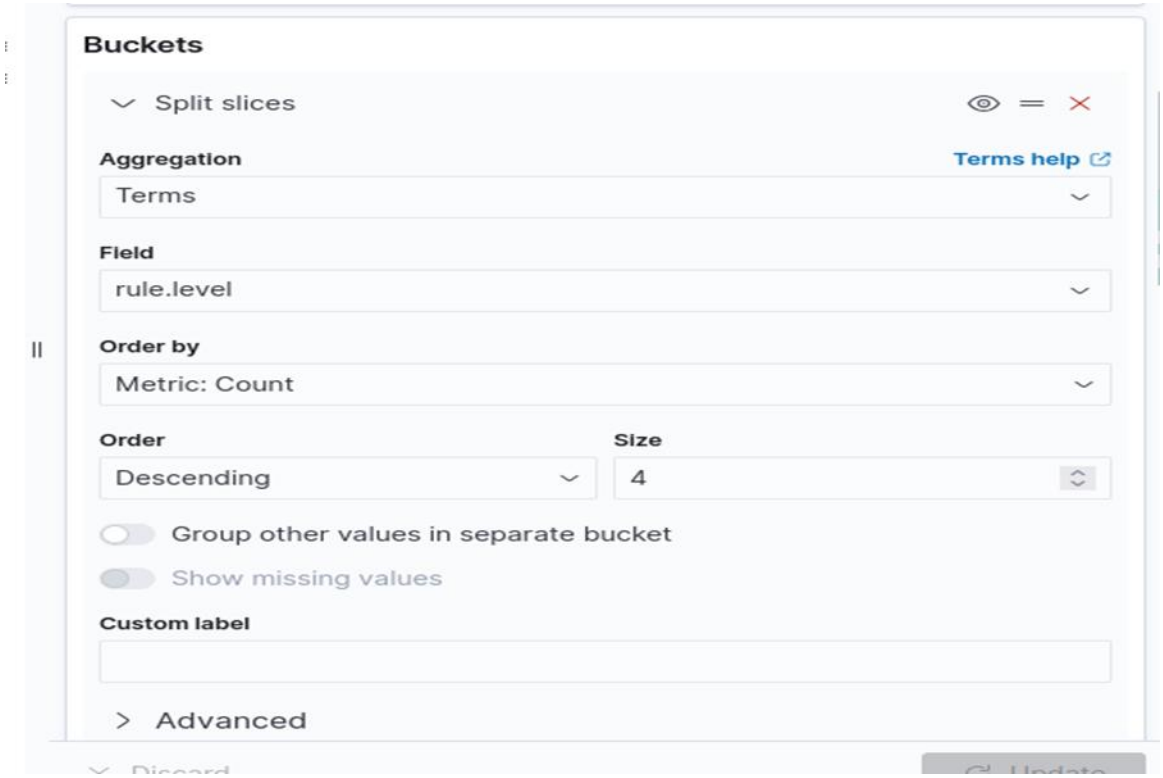

Figure 4: configuring piechart

Bucket → click add backet → split slices →

Aggregation: terms
Field: rule.level
Size: 4

Add filters →sub aggregation → filters
Filter 1: rule.level >= 14 and rule.level <= 15 →critical
Filter 2: rule.level >= 11 and rule.level <= 13 →high
Filter 3: rule.level >= 7 and rule.level <= 10 →medium
Filter 4: rule.level >= 0 and rule.level <= 6 → low

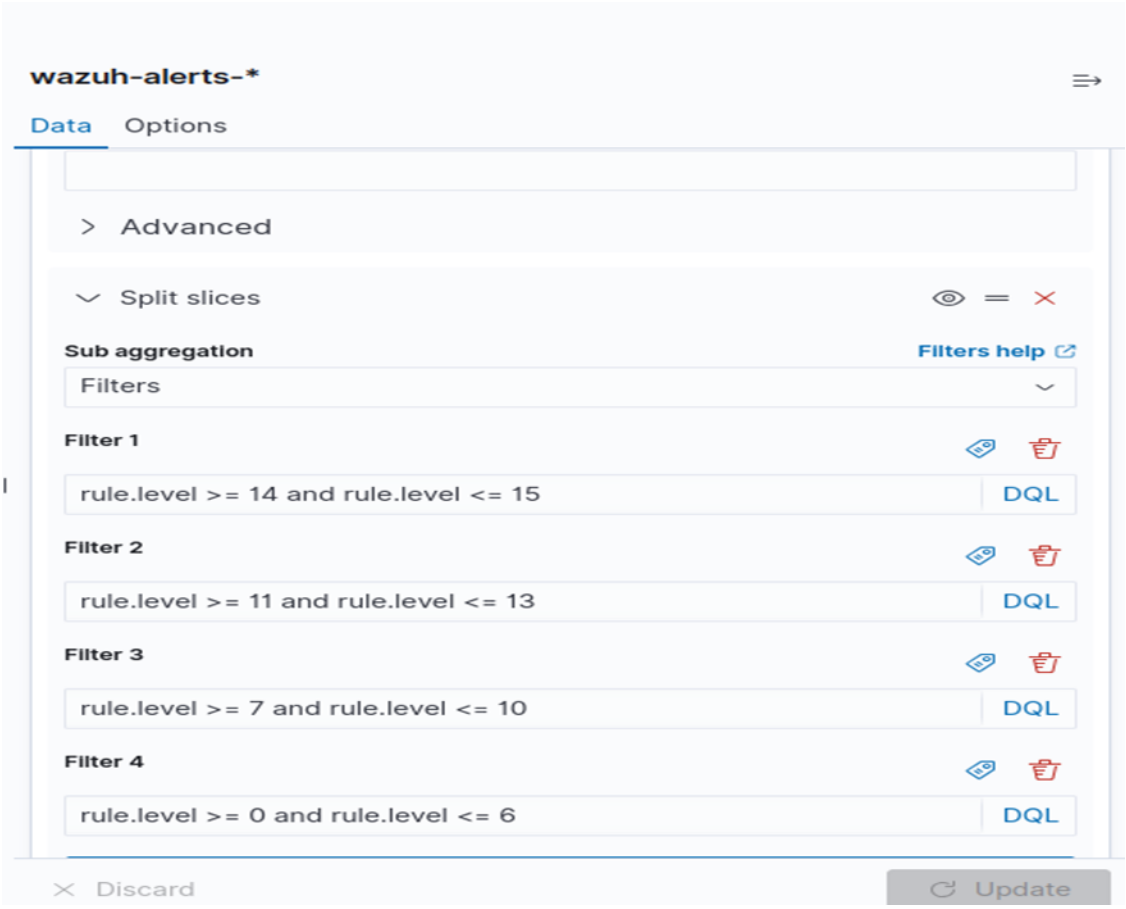

Figure 5: adding filters

Click update and save .

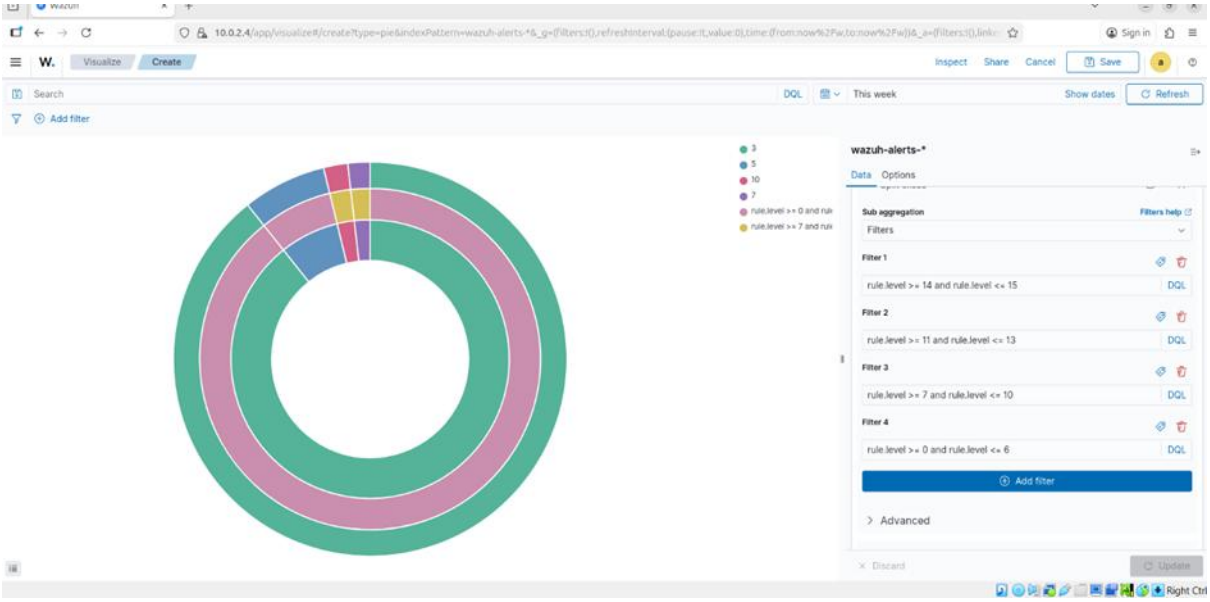

Figure 6: Alert priority distribution

To create dashboard :
Goto dashboard →create new dashboard → click add → select alert priority distribution →save

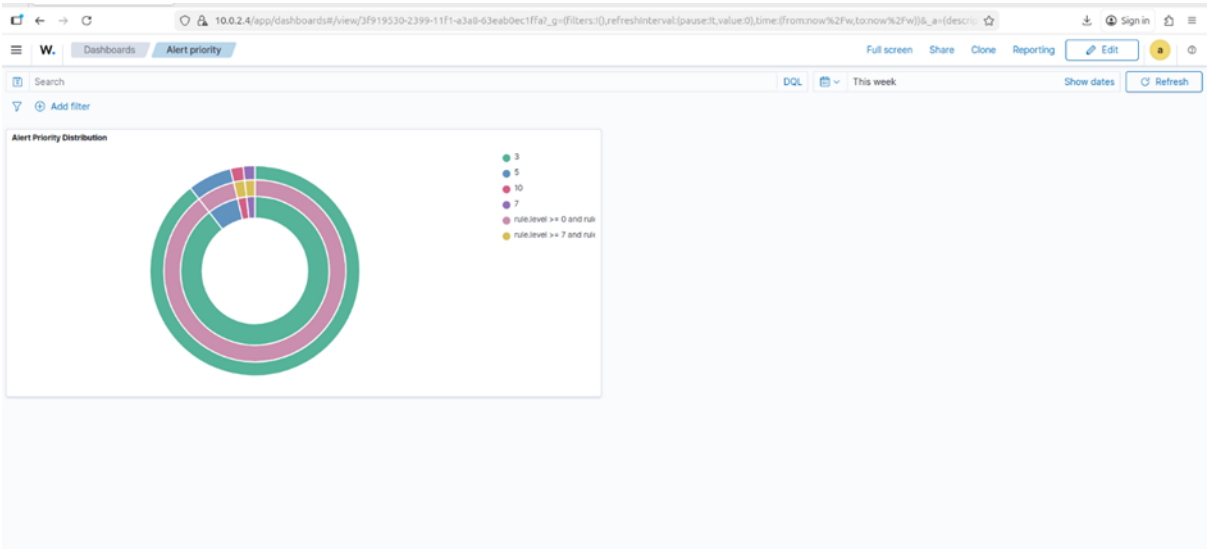

Figure 7: dashboard

Incident Ticket: Draft a ticket in TheHive

- open thehive
  http://localhost:9000
  and log in using username and password
- goto cases → click on new case → add title, severity, description → confirm

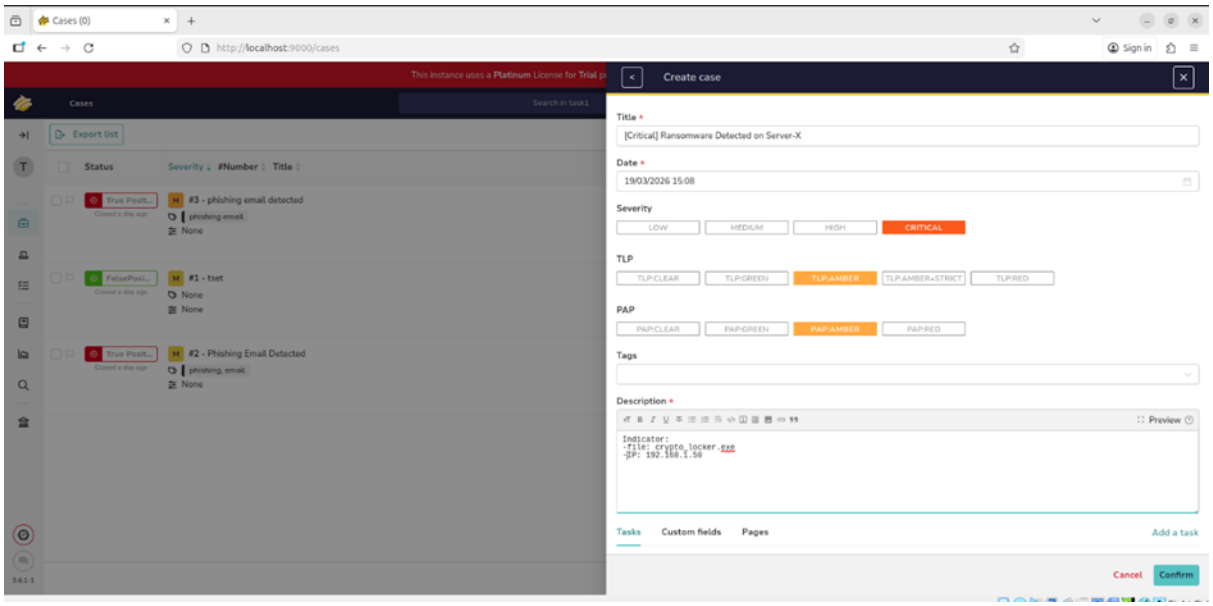

Figure 8: creating new case

- To add assignee: goto task tab → click add task →
  title: analyze ransomware activity
  assignee: soc analyst

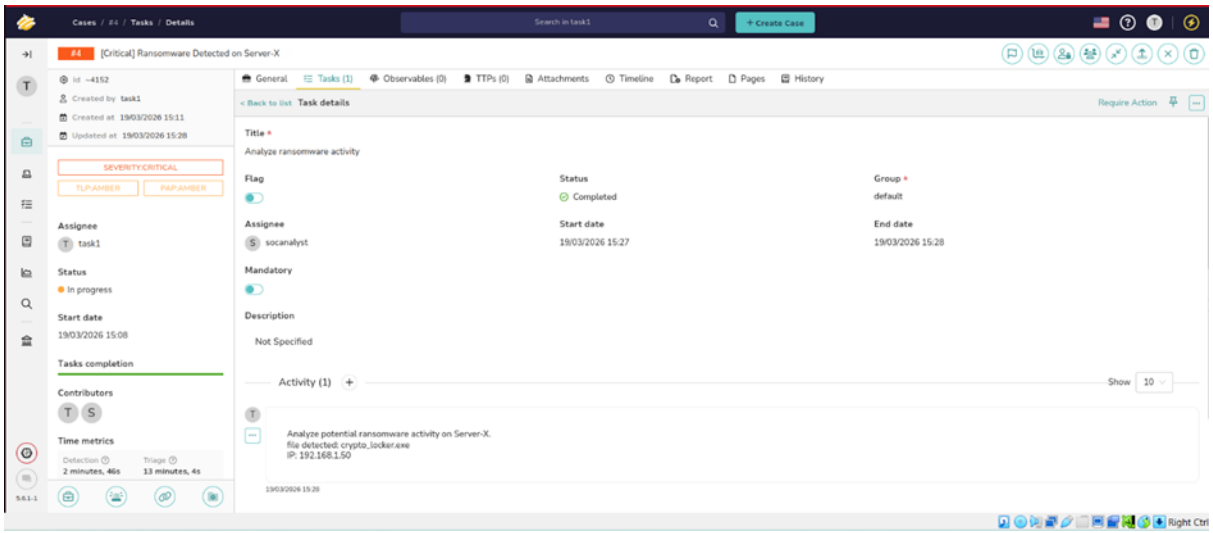

Figure 9: added assignee as soc analyst

- To add observables : goto observables □ add observables □
  (1) Type: filename
  Value: crypto_locker.exe
  Tags: ransomware, malware
  Confirm
  (2) Type: ip
  Value: 192.168.1.50
  Tags: suspicious-ip
  Confirm

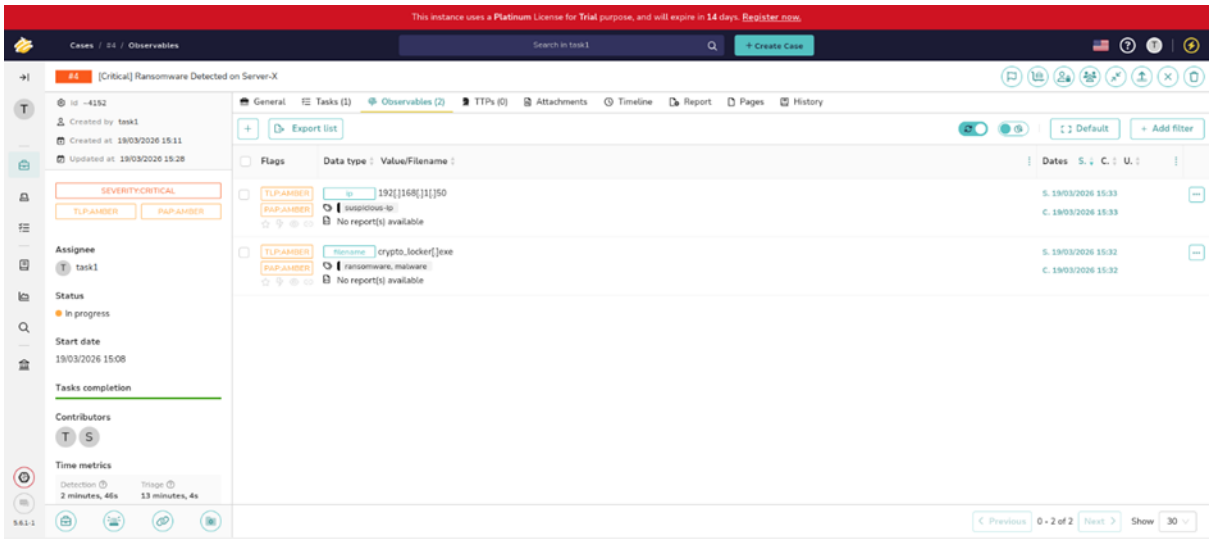

Figure 9: added observables

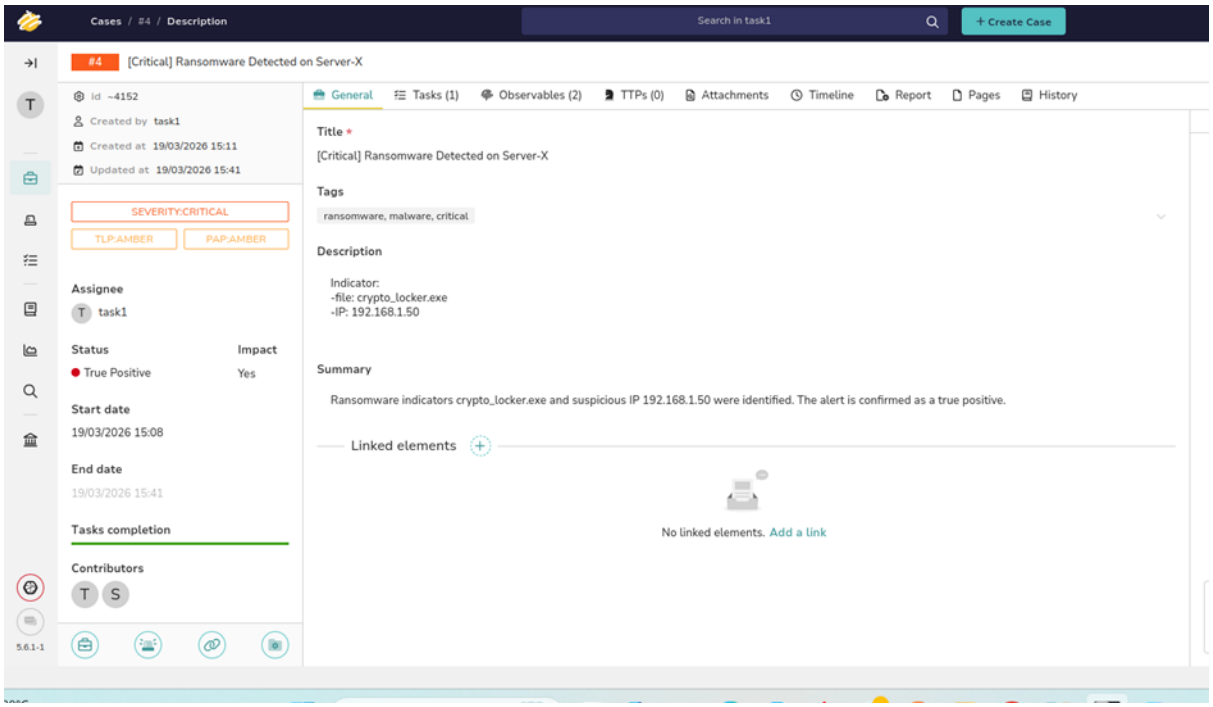

Figure 10: Drafted a ticket in TheHive with given fields

Closed the case in TheHive by marking it as a True Positive and adding a summary of the ransomware indicators.

Escalation Role-Play:
Draft a 100-word email to escalate a Critical alert to Tier 2, summarizing the incident and IOCs:

Subject: Escalation: Critical Ransomware Alert on Server-X

Hi Tier 2 Team,
A critical ransomware alert has been detected on Server-X during routine monitoring.
Initial analysis indicates the presence of a malicious file crypto_locker.exe and suspicious communication with IP address 192.168.1.50.
The alert has been classified as a true positive based on observed indicators.
Immediate investigation is required to determine the extent of compromise and contain potential spread.
The case has been created and documented in TheHive with all relevant details and observables.
Kindly prioritize this incident and proceed with advanced analysis and response actions.

Regards,
SOC Analyst

In the task alert management and incident response simulation provided practical experience in classifying, prioritizing, and escalating security alerts. Using Google Sheets, alerts were mapped to MITRE ATT&CK techniques and scored with CVSS, helping determine their priority. A Wazuh dashboard was created to visualize alert severity, improving situational awareness. A critical ransomware alert was documented in TheHive, including indicators such as malicious files and suspicious IPs, and escalated to Tier 2 for advanced investigation. Drafted a email to escalate a Critical alert to Tier 2.